Designing Constitutional Review Institutions by Simulation

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Abstract

Constitutional review institutions face a recurring design problem: they need enough independence to preserve rights and legal continuity, but enough accountability to avoid becoming unreviewable partisan veto points. We introduce a computational design-search model for comparing constitutional-review institutions under common docket and political assumptions. The model represents appointment systems, court size, tenure, removal, recusal, emergency procedure, voting thresholds, panel and en banc review, cross-checking courts, constitutional councils, legislative overrides, admissibility filters, lower-court splits, repeat-player advantages, and post-decision political response. The current campaign evaluates twenty-two institutional designs across twenty ordinary and adversarial assumption cases. A filed-universe intake model routes cases through review paths and admission filters before merits review, and the reports keep direct outputs such as rights-claimant success, public confidence, lower-court compliance, government noncompliance, precedent durability, emergency downstream effects, and doctrinal depth separate from derived indices. The results do not identify an optimal court. They show how design bundles shift tradeoffs among legal stability, rights protection, emergency-docket abuse, partisan alignment, legitimacy, constitutional conflict, and democratic responsiveness.

Keywords: constitutional review, judicial politics, institutional design, simulation, courts

1 Design Question

The central question is not whether one court design is ideal in isolation. It is how bundles of constitutional-review rules alter institutional tradeoffs when they face the same filed universe, the same political environment, and the same strategic pressures. A court can be independent while losing democratic legitimacy, responsive while failing rights protection, stable while entrenching poor doctrine, or active while creating constitutional conflict. The simulator is built to expose those tensions rather than collapse them into a single institutional label.

This makes the paper a computational institutional-design exercise for judicial politics and comparative public law. The simulator is not intended to forecast any real supreme court. It is a design-search scaffold for asking how appointment regularity, emergency procedure, admission filtering, voting thresholds, override rules, council warnings, cross-court checks, and post-decision compliance strategies change the distribution of outcomes. That framing is especially important because many proposed court reforms are defended one rule at a time, even though their effects depend on the surrounding institutional bundle.

2 Contribution

The paper makes three contributions. First, it treats constitutional-review reform as a design-bundle problem rather than as a sequence of isolated proposals. Court size, tenure, appointment rules, emergency procedure, voting thresholds, review access, and override rules interact, so we compare bundles under shared assumptions instead of ranking individual reforms by intuition. Second, it separates direct model outputs from derived indices. The campaign reports legal stability, rights protection, rights-claimant success, domain-specific claimant success, emergency grant behavior, public confidence, elite acceptance, lower-court compliance, and constitutional conflict before any summary score is used. Third, it provides a reproducible design-search workflow that links scenario definitions, normalized calibration inputs, deterministic seeds, generated reports, and manuscript figures.

These contributions are methodological rather than doctrinal. We do not argue that simulation can settle constitutional design. We argue that explicit simulation can make institutional tradeoffs more inspectable, especially when reforms that look attractive in isolation shift costs into emergency procedure, admission filtering, public legitimacy, interbranch conflict, or post-decision compliance.

3 Claim Boundaries and Interpretation

The manuscript separates empirical claims, synthetic findings, and speculative design recommendations. Empirical claims are limited to source descriptions, normalized calibration guardrails, cited literature, and reproducible project artifacts. They support statements such as which source families are used for plausibility checks, what institutional features are represented in the code, and whether generated artifacts match their manifests. They do not establish causal effects of any reform in the world.

Synthetic findings are outputs of the simulator under stated coding choices, deterministic seeds, and assumption cases. Statements about a scenario reducing shadow-docket abuse, increasing administrative cost, changing lower-court compliance, or moving public confidence are therefore simulation findings, not empirical findings about observed courts. The generated tables and figures are best read as mechanism diagnostics: they show where costs move when a rule bundle changes.

Speculative design recommendations are conditional inferences from those synthetic findings. The paper does not recommend weakening or strengthening courts as an end in itself. The design objective used here is narrower: reduce arbitrary emergency power while preserving rights protection, visible legal reasoning, lower-court settlement, and credible implementation. Reforms that reduce emergency opacity while increasing noncompliance, recusal pressure, or rights failure are not treated as improvements merely because they restrain a court.

4 Theory and Design Space

The model joins three law-and-courts traditions that are often studied separately. The first treats judges as purposive actors whose choices depend on legal goals, policy preferences, colleagues, litigants, and other institutions (Segal and Spaeth, 2002; Epstein and Knight, 1998; Clayton and Gillman, 1999). The second treats constitutional review as an institutional settlement among political actors who face uncertainty, seek insurance against future exclusion, or try to preserve influence through courts (Ginsburg, 2003; Hirschl, 2004; Dixon and Ginsburg, 2017). The third asks how courts preserve authority when they depend on elected officials, lower courts, publics, and repeat players for implementation (Vanberg, 2005; Staton, 2010). A useful simulator for constitutional-review design has to represent all three: judicial choice, institutional allocation, and post-decision compliance.

Comparative constitutional design also makes clear that "judicial review" is not one institution. Strong-form review, weak-form review, pre-enactment review, legislative override, individual complaint, amparo, concrete referral, constitutional council screening, specialized constitutional courts, and ordinary supreme courts expose different actors to different veto points (Tushnet, 2008; Gardbaum, 2013; Sweet, 2000; Brewer-Carías, 2009; Kommers and Miller, 2012). Override and weak-form review scholarship also shows that post-decision political response is part of the institution rather than an afterthought (Hogg and Bushell, 1997; Comella, 2009). That is why this project models access path, review timing, remedy threshold, emergency procedure, and override rule as separate design fields. The relevant unit of comparison is the bundle, not a single reform label.

Emergency procedure is treated separately because it changes both timing and legitimacy. The literature on the U.S. shadow docket emphasizes that orders outside the ordinary merits process can be consequential precisely because they are faster, less reasoned, and less procedurally regular (Baude, 2015; Vladeck, 2023; Kastellec and Taboni, 2026). The simulator therefore separates ordinary merits review from emergency applications, shadow relief, public disagreement, reasons, merits follow-through, and status-quo effects. This makes emergency legality, rights protection, and legitimacy costs visible instead of burying them in an aggregate activity rate.

The design problem also differs from ordinary policy optimization. Constitutional-review institutions produce several goods that can conflict: rights protection, legal stability, responsiveness to democratic majorities, settlement of interbranch conflict, and visible decisional legitimacy. A design that improves one good can undermine another by changing access, delay, information, coalition structure, or implementation incentives. For that reason, the simulator does not search for a single welfare-maximizing court. It asks whether a design bundle moves costs into predictable locations and whether those costs remain visible in direct outputs.

This framing matters for reform debates. A proposal to impose term limits is not only a tenure proposal; it changes vacancy timing, appointment incentives, and the value of partisan capture. A proposal to require reasoned emergency orders is not only a transparency proposal; it changes speed, emergency grant behavior, public disagreement, and merits follow-through. A proposal to add a constitutional council or judicial council is not only a rights-screening proposal; it changes ex ante review, appointment insulation, later judicial workload, and the political salience of invalidation (Garoupa and Ginsburg, 2009). The simulation architecture therefore treats legal form, access path, and political response as jointly produced institutional behavior.

5 Expectations

The simulations are exploratory, but they are not atheoretical. The campaign is organized around five design expectations that can be evaluated in direct outputs rather than only in a final score.

First, regularized appointments should reduce incentives to manipulate appointment timing and should soften public-legitimacy penalties, but the effect should be modest unless the regularized system also changes confirmation control or partisan appointment opportunities. Term limits are therefore expected to improve public confidence and reduce strategic pressure more than they transform rights outcomes.

Second, fragmented or screened appointment systems should reduce partisan alignment and some appointment-manipulation pressure, but they can add administrative cost and may not improve rights protection by themselves. The model therefore expects appointment screening to move legitimacy and alignment more than merits outcomes.

Third, emergency relief without merits follow-through should be a distinct legitimacy risk. Designs that require merits follow-through, reason giving, or full-court referral should reduce shadow-docket abuse and emergency legitimacy risk, but may pay for that improvement through more accelerated merits work or slower emergency handling.

Fourth, panel routing, cross-checking courts, dual supreme courts, and constitutional councils should not be treated as simple activity reducers. They may improve error correction, ex ante screening, or cross-institutional moderation, but they can also raise administrative cost, disagreement, or constitutional conflict if the rules for resolving institutional disagreement are weak.

Fifth, legislative override rules and accountability devices should increase democratic responsiveness only when they do not collapse into repeated override loops, rights carveout evasion, or symbolic compliance. The relevant output is therefore not only whether override is available, but whether it changes practical implementation, rights protection, elite acceptance, and constitutional conflict.

6 Model

Each generated world contains a pool of potential justices, a filed universe of constitutional disputes, and a legislative-output profile. Filed cases vary by legal ambiguity, rights burden, democratic mandate, partisan salience, emergency pressure, requested emergency relief, legislative quality, constitutional-conflict potential, public attention, override pressure, lower-court conflict, lower-court error risk, lower-court split depth, certiorari pressure, strategic plaintiff selection, repeat-player advantage, government noncompliance risk, and recusal incentive pressure. We separate facial challenges, as-applied challenges, election disputes, emergency stay applications, executive-power disputes, administrative-law challenges, structural cases, economic-regulation cases, and rights claims. Legal-domain profiles shift baseline case attributes so rights, election, executive-power, administrative, structural, and economic-regulation cases do not behave as interchangeable labels.

A review-path layer distinguishes paid certiorari, IFP certiorari, discretionary certiorari, direct constitutional complaint, amparo, filtered QPC-style referral, abstract ex ante and ex post review, concrete referral, emergency application, interbranch dispute, and electoral review. Admission filtering turns filed disputes into screened-out matters or merits transfers using SG/CVSG-style signal, amicus intensity, lower-court split depth, split maturity, relists, specialist counsel, vehicle-quality defects, repeat-player advantages, strategic plaintiff selection, and conditional reversal probability. There is no preliminary fixed-size top-case selector before admission; the modeled merits docket is an output of the intake process. The state transition is filed matter $\rightarrow$ petition or referral $\rightarrow$ admission decision $\rightarrow$ merits transfer, emergency order, or screen-out $\rightarrow$ post-decision response.

Each scenario supplies a court design. The design selects reviewers, applies recusal rules, routes cases to full-court or panel review, determines emergency relief procedure, applies voting thresholds, optionally invokes cross-checking or council review, and optionally allows legislative or popular override. Emergency applications carry applicant type, application class, requested relief, response request, full-court referral, status-quo effect, public disagreement risk, and merits follow-through. Emergency orders can also create downstream effects through status-quo disruption, unexplained relief, merits follow-through, and later public disagreement. Strategic actors choose whether to comply, evade, reenact, flood emergency applications, organize override campaigns, or pressure appointments. Their response is split into formal legal response and practical implementation response, including repeal, replacement, narrowed reenactment, weak-form override, amendment, court-curbing, delay, administrative substitution, symbolic compliance, bureaucratic resistance, and open noncompliance.

The model uses shared worlds to make institutional comparisons sharper. Within a run, every scenario sees the same filed matters and legislative-output profile. Differences in outputs therefore come from the institutional rule bundle and its induced strategic responses, not from a different case draw. Across runs and stress cases, the world generator changes appointment capture, emergency pressure, rights risk, democratic mandate, public attention, executive defiance, and imported legislative quality. This gives the campaign both within-world comparison and across-assumption sensitivity.

The current design surface is deliberately broader than the tabled scenarios. Court size may be odd or even; terms may be life, fixed, staggered, renewable, or retirement-age bounded; removal may be impeachment-like, supermajoritarian, or external-review based; recusal may leave replacement, vacancy, or quorum consequences; and remedies may require different thresholds for invalidation, emergency relief, or override. The scenario catalog includes randomized merits panels with en banc correction, mandatory written emergency reasoning, automatic merits follow-up, strong recusal enforcement, jurisdiction-stripping constraints, legislative override windows, constitutional remand, and public-interest litigation filtering. The diagnostics vary assumptions around these bundles rather than presenting every possible combination as if it were substantively meaningful.

That choice keeps the paper focused on institutional mechanisms rather than arbitrary parameter permutations.

An individual simulated dispute moves through the model as follows. First, the filed universe draws a legal domain and case type, then applies the imported legislative-output profile to the case's quality, mandate, rights burden, and conflict potential. A weakly supported, high-salience statute with a high rights burden is therefore not simply "a rights case"; it is a rights case with a particular legislative pedigree and public environment. Second, the access-path module assigns the procedural route. A similar substantive dispute may enter as certiorari, constitutional complaint, amparo, concrete referral, abstract review, emergency application, or interbranch dispute. Third, the admission filter applies path-specific signals. Lower-court conflict, lower-court split depth, SG/CVSG-style support, amicus intensity, split maturity, relists, repeat-player advantage, strategic plaintiff selection, counsel quality, vehicle defects, and conditional reversal probability affect cert-like paths, while complaint and referral paths face different gatekeeping. Fourth, admitted disputes move to merits review or emergency disposition under the scenario's court design. The design determines reviewers, recusals, quorum effects, panel or en banc routing, voting thresholds, reason giving, cross-court or council checks, and emergency merits follow-through. Fifth, the decision creates a formal legal response, a practical implementation response, a precedent-durability value, and an emergency downstream effect. A legislature or executive may repeal, replace, narrow, override, delay, symbolically comply, administratively substitute, or openly defy; lower-court and elite compliance then feed public confidence and constitutional conflict.

The variables in that walkthrough are not fitted coefficients. They are explicit coding choices disciplined by source-range smoke tests, shared-world comparisons, and sensitivity sweeps. This is why the paper reports direct outputs before any score: a design can improve public confidence by reducing opaque emergency relief while leaving domain-specific claimant success nearly unchanged, or it can raise democratic responsiveness while also increasing override loops and conflict. The method is useful only if those movements remain visible rather than being hidden inside a single index.

7 Calibration and Validation

The calibration layer is deliberately framed as a plausibility guardrail rather than as a claim of full validation. Normalized source observations are drawn from the Supreme Court Database, Harvard Law Review Supreme Court statistics, the Supreme Court Shadow Docket Database, and Black and Epstein's recusal data (Supreme Court Database, 2025; Harvard Law Review, 2025; Kastellec and Taboni, 2026; Black and Epstein, 2005). Comparative synthesis rows generated during project research are retained as structured design-context inputs with source names, URLs, confidence labels, validation-use fields, denominator notes, and coverage scopes, but the synthesis memo itself is not treated as an empirical authority. A row-level provenance manifest records the normalized file, source version, source URL, raw-source requirement, transformation script, extraction mode, validation use, and denominator specification for each calibration family. The current calibration run keeps seventeen targets within their assigned ranges; those targets are explicitly classified as strict validation checks, loose calibration checks, proxy sanity checks, mechanism sanity checks, or model-prior checks.

The model distinguishes three categories that should not be blurred. Observed inputs anchor source ranges or qualitative rules. Coding decisions translate legal institutions into parameterized mechanisms. Simulated outputs are exploratory comparisons under those mechanisms. Several quantities remain intentionally loose: SCOTUSblog significant-application counts are not full emergency denominators; raw cross-national invalidation rates are not directly comparable without institutional context; and source tables describing appointment actors or court structure are not treated as direct measures of independence. The generated source-range appendix reports raw source minima, percentiles, medians, maxima, terms, and source keys, while the campaign reports preserve deterministic seeds and scenario definitions for reproduction.

Table 1 reports representative guardrails rather than every source row. It is meant to show the type of check being applied: a current-like design should have plausible docket composition, invalidation, recusal, and emergency-order behavior, while a no-emergency-relief-without-merits design should sharply suppress shadow relief without erasing emergency merits acceleration. "Within range" does not mean the model is causally identified or externally validated. It means the run has not drifted outside ranges that would make the comparison facially implausible for the stated guardrail use.

Table 1: Selected calibration guardrails

Guardrail	Scenario	Use	Observed	Range	Source basis	Status
Merits transfer	Current-like	model prior check	0.498	0.250–0.850	source register; model fallback	within
Invalidation	Current-like	strict validation	0.054	0.000–0.223	SCDB; source range	within
Emergency filings	Current-like	strict validation	0.047	0.000–0.223	shadow database; source range	within
Shadow abuse	Current-like	loose calibration	0.208	0.000–0.657	shadow database; source range	within
Partisan alignment	Commission	proxy sanity check	0.025	0.000–0.657	shadow database; source range	within
Emergency restraint	No-gap	mechanism sanity check	0.001	0.000–0.657	shadow database; source range	within

8 Metrics

The headline metrics are legal stability, rights protection, partisan alignment, shadow-docket abuse, legitimacy, reversal rate, constitutional conflict, and democratic responsiveness. Legal stability is decomposed into precedent stability, statutory stability, interbranch compliance, precedent durability, and lower-court compliance. Emergency legitimacy risk is reported separately from shadow-docket abuse and emergency downstream effect, so opacity, procedure legitimacy, and later disruption are not collapsed into the same quantity as abusive merits-like relief. Rights-claimant success is reported both as an aggregate direct output and by legal domain for rights, structural, election, executive-power, administrative-law, and economic-regulation cases.

The reported directional score is only a reading aid. It averages a stability/rights score, a legitimacy/control score, rights-claimant success, precedent durability, lower-court compliance, elite acceptance, and administrative feasibility. Direct outputs such as certiorari admission, lower-court split depth, strategic plaintiff selection, repeat-player advantage, government noncompliance, recusal incentive pressure, emergency downstream effect, rights-claimant success, doctrinal depth, remedial breadth, lower-court compliance, elite acceptance, public confidence, and coalition fragmentation are reported separately and should not be treated as interchangeable components of one welfare function.

Figure 1 illustrates why the domain-specific claimant-success measures are kept separate. Designs can look similar on aggregate rights protection while moving success rates differently for rights, structural, election, executive-power, administrative-law, and economic-regulation disputes.

	Rights	Struct.	Elect.	Exec.	Admin.	Econ.
Current-like	0.39	0.24	0.67	0.21	0.17	0.13
18-year terms	0.37	0.25	0.55	0.18	0.16	0.15
Written reasons	0.36	0.24	0.53	0.16	0.14	0.14
Auto merits	0.44	0.28	0.67	0.24	0.20	0.17
No merits gap	0.40	0.24	0.60	0.18	0.17	0.13
Recusal enforce.	0.39	0.25	0.55	0.18	0.18	0.14
Random panels	0.38	0.25	0.52	0.16	0.17	0.14
Public-interest	0.44	0.26	0.56	0.19	0.17	0.16

Domain-specific claimant-success rates; darker cells are higher.

Figure 1: Domain-specific rights-claimant success for selected institutional designs. The aggregate rights-protection score can hide domain shifts, especially for election, structural, executive-power, administrative-law, and economic-regulation cases.

9 Campaign Results

The current campaign compares twenty-two institutional scenarios across twenty assumption cases: the original broad cases, sensitivity sweeps for appointment capture, emergency pressure, rights risk, and democratic mandate, plus adversarial stress cases for appointment timing, emergency-application flooding, override evasion, recusal pressure, and court-expansion retaliation. The reported run used 80 runs per case, 64 filed matters per run, seed 20260501, and an imported legislative-output profile from a separate legislative simulator report contract.

Figure 2 plots public confidence against constitutional conflict for selected designs, with marker shade representing shadow-docket abuse. The figure is diagnostic rather than normative: movement toward lower conflict and higher confidence is usually attractive, but the figure intentionally leaves rights protection and democratic responsiveness outside the two-dimensional display.

Table 2 should be read as a clustered comparison, not as a tournament table. Differences below .010 in the directional score are treated as substantively indistinguishable, and even larger differences should be interpreted through the direct outputs before drawing any design conclusion.

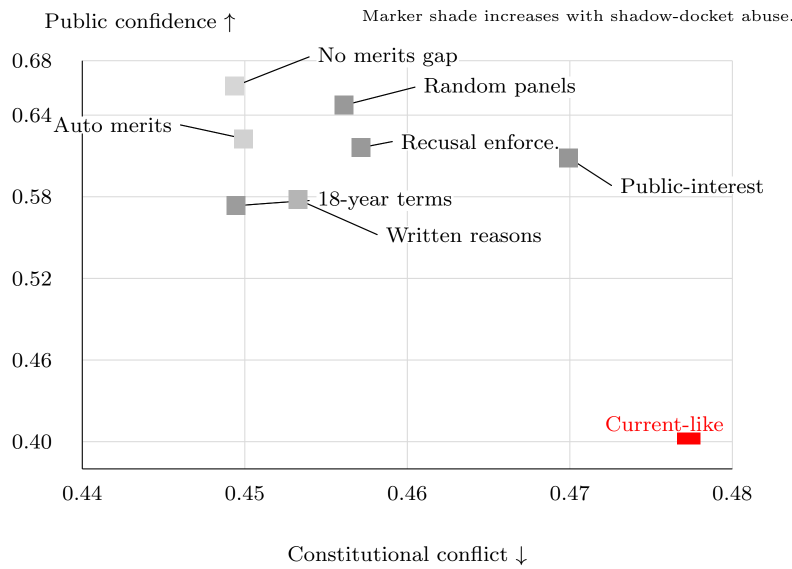

Figure 2: Public-confidence and constitutional-conflict tradeoff for selected institutional designs. Marker shade increases with shadow-docket abuse, so the figure separates legitimacy and conflict from emergency-docket opacity instead of folding them into one score.

Table 2: Selected v2 campaign averages, grouped by emergency-power and rights-protection profile

Scenario	Rights	Shadow	Emerg. risk	Downstream	Gov. noncomp.	Lower ct.	Prec. dur.	Public	Score
Stylized current U.S.-like court	0.649	0.302	0.329	0.151	0.198	0.477	0.702	0.406	0.562
18-year staggered terms	0.643	0.086	0.242	0.082	0.182	0.500	0.747	0.573	0.576
Mandatory written emergency reasoning	0.640	0.050	0.250	0.093	0.183	0.499	0.764	0.578	0.571
Automatic merits follow-up	0.654	0.011	0.230	0.082	0.182	0.505	0.731	0.623	0.571
No emergency relief without merits review	0.651	0.005	0.233	0.068	0.178	0.507	0.764	0.661	0.578
Independent recusal enforcement with substitutes	0.647	0.090	0.250	0.085	0.188	0.496	0.752	0.616	0.569
Randomized merits panels with en banc correction	0.645	0.089	0.247	0.085	0.184	0.498	0.759	0.647	0.569
Constitutional remand before invalidation	0.643	0.092	0.255	0.087	0.191	0.500	0.797	0.640	0.566
Public-interest litigation filter	0.658	0.093	0.261	0.088	0.195	0.491	0.755	0.609	0.567

The stable front-line cluster contains several emergency-procedure and process-regularity reforms that preserve rights protection while lowering shadow-docket abuse or emergency downstream effects. Mandatory written emergency reasoning, automatic merits follow-up, no-relief-without-merits review, strong recusal enforcement, randomized panels with en banc correction, constitutional remand, and public-interest filters are therefore compared as different ways to reduce arbitrary emergency power rather than as a simple ladder from weak courts to strong courts. The stylized current U.S.-like court can have substantial rights protection in this run while still producing higher shadow-docket abuse and emergency legitimacy risk. That is the central tradeoff, not a claim that one directional score wins.

Table 3 is the audit table for the concern that formal court design might overweight the model. It shows that the reported outputs now expose litigation-pipeline and downstream-enforcement

Table 3: Litigation-pipeline and downstream-enforcement diagnostics

Scenario	Cert admit	Split	Plaintiff sel.	Repeat player	Recusal press.	Gov. noncomp.	Emerg. downstream	Prec. dur.
Current-like	0.566	0.589	0.486	0.260	0.469	0.198	0.151	0.702
18-year terms	0.568	0.589	0.486	0.260	0.477	0.182	0.082	0.747
Written emergency reasons	0.570	0.589	0.486	0.260	0.476	0.183	0.093	0.764
Automatic merits follow-up	0.569	0.589	0.486	0.260	0.477	0.182	0.082	0.731
No emergency merits gap	0.587	0.589	0.486	0.260	0.491	0.178	0.068	0.764
Strong recusal enforcement	0.587	0.589	0.486	0.260	0.497	0.188	0.085	0.752
Random merits panels	0.586	0.589	0.486	0.260	0.619	0.184	0.085	0.759
Jurisdiction-stripping constraints	0.585	0.589	0.486	0.260	0.477	0.193	0.084	0.750
Legislative override window	0.581	0.589	0.486	0.260	0.477	0.194	0.084	0.751
Constitutional remand	0.611	0.589	0.486	0.260	0.482	0.191	0.087	0.797
Public-interest filter	0.621	0.589	0.486	0.260	0.482	0.195	0.088	0.755

quantities directly: certiorari admission, lower-court split depth, strategic plaintiff selection, repeat-player advantage, recusal incentive pressure, government noncompliance, emergency downstream effect, and precedent durability. Those values are synthetic diagnostics, but they force the manuscript to show whether a formal design improvement is actually operating through intake, lower-court settlement, compliance, or emergency-order incentives.

Table 4 reports named-prior uncertainty bands from the parameter sweep. The overlap among the leading designs is the main point. The model is more informative about tradeoff patterns, such as emergency-restraint reducing shadow abuse and dual-court designs increasing conflict under the current disagreement rule, than about fine ordinal rankings among close scores.

Table 4: Campaign scores and uncertainty bands for selected designs

Scenario	Campaign score	Score 5/50/95	Shadow 5/50/95	Conflict 5/50/95
18-year terms	0.576	0.533/0.605/0.617	0.044/0.062/0.134	0.334/0.373/0.559
Written emergency reasons	0.571	0.527/0.600/0.612	0.018/0.029/0.086	0.331/0.374/0.573
Emergency merits follow-up	0.571	0.527/0.601/0.611	0.003/0.004/0.026	0.338/0.373/0.560
No emergency merits gap	0.578	0.530/0.604/0.615	0.001/0.001/0.013	0.334/0.373/0.563
Current-like	0.562	0.514/0.590/0.606	0.171/0.228/0.432	0.344/0.388/0.607
Strong recusal enforcement	0.569	0.522/0.598/0.610	0.048/0.064/0.143	0.340/0.382/0.578
Constitutional remand	0.566	0.519/0.592/0.605	0.047/0.066/0.141	0.348/0.387/0.594

Figure 3 makes the emergency-procedure result visible. Similar rights and stability averages can coexist with very different emergency profiles, so the manuscript reports shadow-docket abuse, emergency legitimacy risk, and public confidence separately.

10 Mechanism-Level Summary

The scenario table names institutional bundles. Table 5 reports a second view: weighted paired contrasts from the mechanism-ablation report. Each row compares a current-like baseline to one institutional change across the same assumption cases. These are not causal estimates from observed courts, and they are not a substitute for the scenario table. They are a compact check on whether the named mechanisms move the outputs in the directions implied by the expectations.

The contrast table reinforces the main interpretation. Emergency restraint, written emergency reasoning, automatic merits follow-up, and strong recusal enforcement move shadow-docket and downstream-emergency diagnostics more directly than appointment or size reforms. Randomized panels, constitutional remand, public-interest filtering, jurisdiction-stripping constraints, and legislative override windows have more mixed profiles because they affect access, correction, compliance,

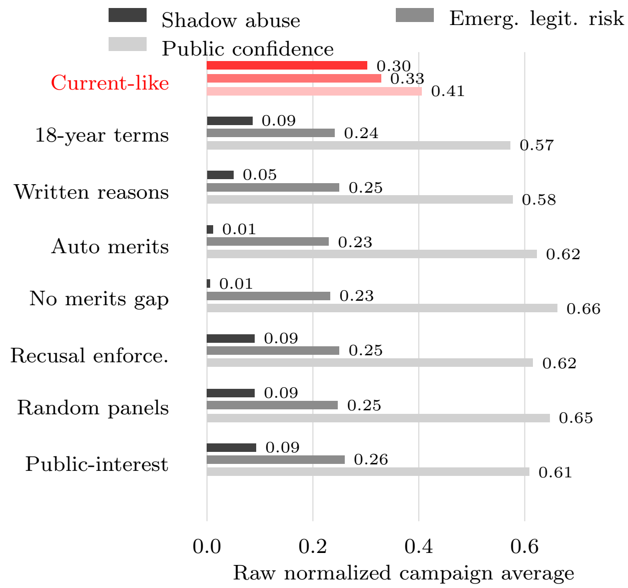

Figure 3: Emergency-docket and public-confidence profile for selected institutional designs. The comparison highlights why emergency procedure is modeled directly: designs that are similar on rights protection and legal stability can differ sharply on shadow-docket abuse, emergency legitimacy risk, and public confidence.

Table 5: Mechanism-level paired contrasts against the current-like design

Mechanism	Score	Rights	Shadow	Lower ct.	Gov. noncomp.	Emerg. downstream	Prec. dur.	Cost
Emergency restraint	+0.018	+0.001	-0.329	+0.033	-0.018	-0.095	+0.064	+0.095
Mandatory written emergency reasoning	+0.010	-0.014	-0.279	+0.024	-0.012	-0.065	+0.066	+0.092
Automatic merits follow-up	+0.011	+0.006	-0.318	+0.030	-0.015	-0.080	+0.033	+0.112
Strong recusal enforcement	+0.009	-0.006	-0.236	+0.022	-0.013	-0.074	+0.053	+0.100
Randomized merits panels	-0.008	+0.003	+0.003	-0.002	-0.002	+0.003	+0.005	+0.053
Constitutional remand	-0.007	-0.005	+0.001	+0.005	+0.005	+0.001	+0.049	+0.112
Public-interest litigation filter	-0.006	+0.009	+0.004	-0.006	+0.009	+0.004	+0.006	+0.040
Legislative override window	-0.007	+0.004	+0.002	-0.004	+0.008	+0.002	+0.003	+0.036
Jurisdiction-stripping constraints	-0.003	+0.002	+0.002	-0.004	+0.012	+0.001	+0.000	+0.027

or post-decision response rather than simply reducing court activity. These are synthetic mechanism contrasts, not observed causal estimates. Their use is to identify which mechanisms reduce arbitrary emergency power while preserving rights protection and which mechanisms merely move costs into noncompliance, administrative burden, or lower-court instability.

11 Limitations

The model is strongest as a disciplined comparison of institutional mechanisms, not as a predictive empirical model. It does not estimate causal effects from observed constitutional systems. It uses source-backed ranges, comparative institutional coding, and deterministic scenario campaigns to identify where rule bundles create tradeoffs that would otherwise be difficult to see. The strongest next validation steps are to automate raw-data refresh for SCDB, Harvard Law Review statistics, shadow-docket data, and recusal data; convert named priors into sampled prior distributions; add actor learning across repeated outcomes; calibrate review-path-specific admission denominators against raw certiorari, amparo, constitutional complaint, QPC, abstract-review, and concrete-referral data; and benchmark the legislative import contract against raw legislative simulator runs rather than only aggregate campaign CSVs.

A Institutional Parameter Provenance

Table 6: Illustrative provenance map for modeled institutional features

Feature family	Model representation	Provenance status
Appointment and tenure	Appointment method, confirmation threshold, vacancy deadlock risk, term renewability, retirement age, replacement timing	Comparative institutional coding and scenario assumptions
Review access	Access path, petition type, admission score, lower-court split depth, strategic plaintiff selection, repeat-player advantage, merits transfer, screen-out	Normalized source-register rows and modeled admission rules
Emergency procedure	Application class, relief request, response request, full-court referral, reason giving, status-quo effect, merits follow-through, downstream effect	Shadow-docket source tables plus scenario rules
Decision rule	Voting threshold, panel/en banc routing, quorum failure, recusal consequence, per-remedy thresholds	Court-design coding and institutional assumptions
Post-decision politics	Override attempt, rights carveout, formal response, practical implementation, government noncompliance, open defiance	Comparative public-law coding and strategic actor assumptions
Outputs	Stability, rights protection, claimant success, legitimacy, conflict, responsiveness, precedent durability, lower-court compliance	Simulated outcomes with calibration guardrails

B Model Mechanics

This appendix records the main state machine and scoring formulas so the manuscript does not treat the simulator as a black box. Each run draws a filed universe from a shared world seed. For each scenario, the same filed matters are reviewed by a court process with a scenario-specific design and a scenario-specific random stream. Let i index filed matters and s index institutional scenarios.

Admission begins with a path-specific petition probability:

$$\begin{aligned} P_{is} = \mathrm{clamp}_{[0,1]}(b_{\mathrm{petition}} &+ .08\,\mathrm{conflict}_i + .08\,\mathrm{attention}_i + .06\,\mathrm{SG}_i \\ &+ .06\,\mathrm{splitDepth}_i + .04\,\mathrm{repeatPlayer}_i + .04\,\mathrm{plaintiffSelection}_i - .22\,\mathrm{vehicleDefect}_i). \end{aligned}$$

Conditional on a petition or referral, the admission score combines certiorari pressure, lower-court conflict, lower-court error, lower-court split depth, SG/CVSG signal, amicus intensity, split maturity, relist signal, specialist counsel, repeat-player advantage, strategic plaintiff selection, conditional reversal probability, access-path adjustment, institutional gatekeeping, vehicle defects, and vacancy deadlock. Merits transfer is then path-specific: emergency applications transfer through the application's merits-follow-through probability, complaints and amparo transfer at a lower rate, and ordinary referrals or cert-like paths transfer at a higher baseline.

After admission, the court process chooses strategic-actor response, emergency status, initial merits review, panel or en banc routing, recusals, quorum, votes, emergency order, invalidation, auxiliary-review outcome, override attempt, and post-decision implementation response. The direct metrics are case averages over the filed universe. For example, rights-claimant success is one when a high-rights-burden claimant receives invalidation or shadow relief and partial when a constitutional remand preserves a claimant's route without immediate invalidation; public confidence is legitimacy minus penalties for emergency legitimacy risk, partisan alignment, appointment-pressure campaigns, and court-curbing response; constitutional conflict rises with conflict potential, executive defiance, lower-court conflict, accumulated state conflict, government noncompliance, emergency downstream effect, invalidating high-mandate laws, shadow relief, cross-court disagreement, override attempts, and repeated override loops.

The directional score is intentionally secondary:

$$\begin{aligned} \mathrm{DirectionalScore}_s = \mathrm{mean}(&\mathrm{StabilityRights}_s, \mathrm{LegitimacyControl}_s, \mathrm{ClaimantSuccess}_s, \\ &\mathrm{PrecedentDurability}_s, \mathrm{LowerCourtCompliance}_s, \mathrm{EliteAcceptance}_s, \\ &1 - \mathrm{AdministrativeCost}_s). \end{aligned}$$

The stability-rights component averages legal stability, precedent durability, lower-court compliance, rights protection, reversal restraint, and conflict restraint. The legitimacy-control component averages legitimacy, democratic responsiveness, independence-accountability balance, partisan-alignment restraint, shadow-abuse restraint, emergency-legitimacy-risk restraint, emergency-downstream restraint, government-noncompliance restraint, recusal-pressure restraint, and strategic-pressure restraint. Because those choices are normative, the paper reports the direct outputs separately and uses the directional score only as a reading aid. Score differences below .010 are treated as indistinguishable in the manuscript's substantive interpretation.

Data Availability Statement

The simulator, normalized calibration tables, frozen legislative-output fixtures, deterministic seeds, scenario definitions, generated reports, figure-generation scripts, standalone figure exports, source-

Table 7: Representative model weights and decision rules

Component	Representative rule	Interpretation
Admission	0.18 + .22 cert pressure +.17 lower-court conflict +.10 lower-court error +.10 split depth +.15 SG/CVSG signal +.10 amicus signal +.08 split maturity +.07 relist signal +.05 specialist counsel +.06 repeat-player signal +.05 strategic-plaintiff signal +.08 conditional reversal −.16 vehicle defect, plus path adjustment	Admission is an institutional screen, not a fixed docket quota.
Voting	Legal concern receives weight .42; rights concern uses rights burden, justice rights sensitivity, and weight .44; democratic mandate, legislative quality, institutionalism, emergency deference, repeat-player pressure, precedent stability, and accountability pull reduce or increase invalidation propensity.	Merits votes mix legal, rights, strategic, institutional, and emergency-deference considerations.
Emergency abuse	Open emergency rule base .32, no-merits penalty .22, shadow stay .24, written reasons −.10, automatic merits follow-up −.12, merits acceleration −.07, partisan salience .12, ambiguity .08.	Shadow-docket abuse is driven by opacity and merits-like relief without ordinary merits process.
Legitimacy	Base .48, public attention .16, legislative quality .10, merits reason-giving .18, recusal discipline up to .16, council warning .06, shadow abuse −.24, partisan alignment −.20.	Public legitimacy rewards visible process and penalizes opaque or partisan-looking intervention.
Conflict	Conflict potential .38, executive defiance .18, lower-court conflict .08, conflict load .12, legislative defiance .10, government noncompliance, emergency downstream effect, invalidating high-mandate laws .14, shadow relief .12, cross-court disagreement .16, override attempt .06.	Constitutional conflict is cumulative and rises when legal disagreement becomes interbranch disagreement.
Override	Attempt pressure combines override pressure .34, democratic mandate .30, public attention .18, conflict load .12, legislative defiance .12, override adaptation .10, and rights-burden penalty −.24.	Accountability devices affect implementation only after invalidation and can generate rights carveouts or repeated override loops.

audit file (`paper/source-audit.csv`), source-provenance manifests, and report manifests are maintained as a reproducible project. Public repository identifiers are withheld in this anonymous review manuscript. The intended replication package includes source code, build instructions, normalized input data, generated campaign artifacts, generated figure fragments, standalone figure files, manifest hashes, and documentation distinguishing observed inputs, coding choices, model assumptions, and simulated outputs. For final publication, the package should be deposited in the Journal of Law and

Courts Dataverse or another repository linked from the JLC Dataverse rather than being available only on request.

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